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METHOD ARTICLE

Developing a place-based audio guide to support nature engagement: Intervention content and programme theory

[version 1; peer review: 2 approved with reservations, 1 not approved]

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Abstract

Nature-based interventions may support health and wellbeing but contact with natural environments is in itself not always enough – the structure and quality of engagement with nature also shape outcomes. This is particularly relevant for people with mobility impairments, who often face physical and social barriers to nature engagement. This study aimed to develop and describe a place-based audio guide for accessible nature visits and to refine the intervention prior to further evaluation. The guide was developed within a complex intervention framework. Development was iterative, drawing on theory, empirical literature, evidence on mobility-related barriers to nature access and engagement and a feasibility pilot. The final guide consisted of three audio tracks delivered at predefined stops along an accessible nature route. The guide was structured around three components: present-moment awareness, biological interconnectedness and nature knowledge. Pilot findings suggested that the guide could support attentional orientation, sensory engagement and reflective meaning-making, while also identifying challenges related to pause uncertainty, sensory interference, route structure and prompt–context mismatch. These findings informed refinement of the content, delivery options and programme theory. This study provides a transparent, theory- and feasibility-informed basis for further process and outcome evaluation of a guided nature-based intervention.

Keywords

Nature-based health interventions, nature engagement, nature connection, audio guide, accessibility, mobility impairment, complex intervention, design principles, guided nature visits

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Introduction

Nature-based interventions have shifted beyond a sole focus on contact with nature toward more structured ways of supporting how participants attend to and interact with natural environments (Sheffield et al., 2022; Wood et al., 2025). Central to this shift is the concept of nature connection, understood as how individuals perceive and experience their relationship with the natural world (Tam, 2013; Martin et al., 2020). Nature connection has been linked to wellbeing, restoration and pro-environmental behaviour, and interventions have therefore increasingly incorporated guidance intended to support this relationship (Barragan-Jason et al., 2023; Martin et al., 2020).

At the same time, evidence suggests that the effects of guided nature-based approaches are heterogeneous (Sheffield et al., 2022). This may reflect variation in how participants respond to such support, including differences in prior experience, preferences and the way opportunities for interaction are structured (Pasanen et al., 2018; Sheffield et al., 2022). The effects of such interventions therefore depend as much on how the intervention is designed and delivered as on the environment itself, making them relevant to develop and evaluate within a complex intervention framework (Skivington et al., 2021; Madsen et al., 2026).

This is particularly important when developing interventions for population groups who face barriers to meaningful engagement with natural environments. People with physical impairments represent a substantial population – approximately 15% of the world's population live with some form of disability, with mobility disabilities among the most common (WHO, 2023) – who may encounter both physical and experiential barriers to nature engagement while also experiencing broader inequalities in health and wellbeing (Bekke-Hansen et al., 2023; Corazon et al., 2019; Stigsdotter et al., 2018). There is therefore a need for approaches that can support engagement with natural environments when independent exploration is limited.

One such approach is simple guidance that directs attention to features of the surrounding environment during a nature visit (McEwan et al., 2019). The present study addresses this need through the development of a place-based audio guide designed to support structured engagement during an accessible nature visit for individuals with mobility impairments. The intervention consists of three audio-guided stops along an accessible nature route, where participants are invited to engage with the environment through sensory attention, reflective prompts and place-based nature knowledge. The audio guide was initially developed within the Move Green project and was refined through a pilot study. It was subsequently incorporated into the Horizon Europe project RESONATE, which includes a two-arm randomized controlled intervention comparing audio-guided engagement with unguided nature visits (Bekke-Hansen et al., in review; White et al., 2026).

Aim

The aim of this study was to develop and describe a place-based audio guide designed to support nature engagement.

Methods

Reporting and methodological standards

The audio guide was developed within a complex intervention framework, informed by the Medical Research Council guidance for developing and evaluating complex interventions (Skivington et al., 2021). The documentation of the development process was guided by the Guidelines for Intervention Development Studies (GUIDED) checklist (Duncan et al., 2020), and the final audio guide is described using the Template for Intervention Description and Replication (TIDieR) checklist (Hoffmann et al., 2014). Both checklists are provided in the supplementary materials. Reporting of the qualitative pilot component embedded within the development study was informed by relevant items from the Standards for Reporting Qualitative Research (SRQR) (O'Brien et al., 2014), to support transparent reporting of the qualitative design, data generation, analysis, and interpretation.

Evidence inputs

The work on developing the engagement guide followed an iterative process, drawing on the following inputs:

- Synthesis of theory and empirical evidence on engagement interventions in natural environments, nature connection, restorative environment frameworks, broader nature–health theories, and the barriers people with mobility impairments may face in accessing and engaging with natural environments.

- Feedback from a pilot study assessing the fit between the intervention setting and the audio guide, as well as the clarity, pacing and prompt-context fit of the guide.

Figure 1 illustrates the development process of the guide included in this paper.

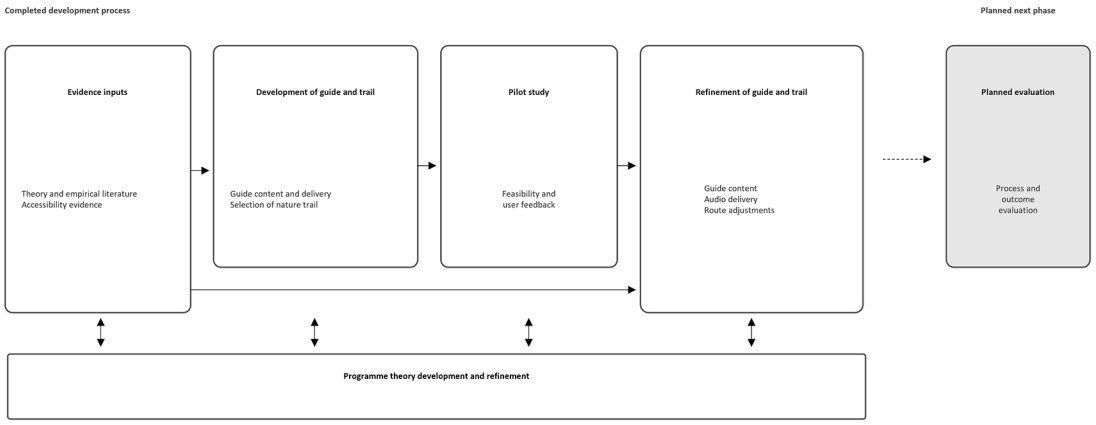


Figure 1. Overview of the iterative development process of the place-based audio guide.

Theories and empirical findings informing the development of the audio guide content

The audio guide was informed by complementary theoretical and empirical perspectives relevant to how engagement with natural environments may be structured and supported. These perspectives were particularly relevant to attention, emotional regulation, nature connection and place-based engagement.

Attention-related perspectives were particularly important in informing the guide. Attention Restoration Theory emphasizes the restorative potential of effortless attention in compatible natural environments (Kaplan, 1995). Related studies suggest that present-moment attention to sensory experience may support sensory noticing and attentional engagement and may contribute to both restorative and connection-related experiences (Lymeus et al., 2018; Macaulay et al., 2022; Nisbet et al., 2019; Richardson et al., 2022). Together, these findings supported the inclusion of prompts designed to direct attention toward sensory experiences in the surrounding environment.

A second perspective that influenced the development of the guide concerned emotional regulation during nature visits. Stress Reduction Theory posits that natural environments can support stress reduction and positive affective responses (Ulrich et al., 1991). This is complemented by research demonstrating that contact with nature may be associated with emotion regulation and may contribute to reduced stress and improved well-being (Vitale & Bonaiuto, 2024). This perspective informed the use of calming, low-demand prompts intended to promote feelings of safety and tranquillity.

A related body of literature concerned nature connection. Nature connection has been described as a multidimensional relational construct and has been linked to emotional, cognitive and experiential aspects of human–nature relationships (Tam, 2013; Martin et al., 2020). It has been suggested that nature connection may partly underlie the beneficial effects of nature-related experiences by supporting adaptive emotion regulation processes and well-being processes more generally (Bakir-Demir et al., 2021; Vitale & Bonaiuto, 2024; Pritchard et al., 2020). As a result, frameworks to support nature connection through engagement have been developed. *The Five Pathways to Nature Connection* suggest that engagement with nature may be structured through pathways such as sensory contact, emotion, meaning, beauty and compassion (Lumber et al., 2017), and this framework helped shape the guide's structure. An additional aspect of nature connection, the feeling of "oneness" with the natural world has a more explicitly relational and existential quality (Mayer & Frantz, 2004). This resonates with biophilic and eco-existential perspectives that further informed the guide by highlighting human embeddedness within broader ecological systems (Kellert & Wilson, 1993; Wilson, 1984; Passmore & Howell, 2014; Jordan & Hinds, 2016). These perspectives suggest that awareness of interdependence with the natural world may contribute to experiences of meaning, authenticity and connectedness, which are often associated with eudaimonic wellbeing (Passmore & Howell, 2014; Jordan & Hinds, 2016; Pritchard et al., 2020). These theoretical orientations were used as perspectives that supported the inclusion of narratives and metaphors related to biological interconnectedness.

Finally, the guide was informed by perspectives on place attachment, which highlight how personal experiences, social connections, and cultural or historical narratives may contribute to attachment to a specific place and, in turn, to wellbeing-related experiences (Basu et al., 2020; Scannell & Gifford, 2010). This perspective was relevant alongside research on disconnection from nature and barriers to engaging with nature, which suggests that engagement with natural environments may be restricted by both limited familiarity and various barriers (Beery et al., 2023; Gittins et al., 2023).

For people with mobility impairments, such barriers can be both physical and immediate, directly restricting opportunities for preferred activities such as close sensory contact and deeper involvement with natural settings (Corazon et al., 2019). In this context, place-based knowledge was included as an additional route to engagement, supporting curiosity, orientation and meaning-making when physical immersion was limited.

Together, these perspectives helped shape the audio guide's three core components: present-moment awareness, biological interconnectedness and nature knowledge.

Core components

The three components, which are described below, reflect different ways of inviting engagement with the environment.

Present-moment awareness

This component includes attentional guidance, directing attention to breathing, bodily awareness and sensory experiences stemming from the surrounding environment. This is done to support present-moment bodily awareness and awareness of the individual in the environment, allowing attention to experiences to unfold without judgment. This is in accordance with existing nature-integrated, mindfulness-based approaches and the contact dimension entailed in the *Five Pathways* (Lymeus et al., 2018; Lumber et al., 2017; Macaulay et al., 2022).

Biological interconnectedness

Biological interconnectedness was included to introduce narratives and metaphors illustrating humans' embeddedness within natural systems, drawing on examples such as respiration and photosynthesis as well as ecological cycles of air, water, life and death. This component presents biological interdependence between individuals and their environment through symbolic imagery linking human and natural processes. Strengthening awareness of interdependence has been described as a means of enhancing relational closeness between people and has been theorized to have a similar relevance for human–nature relationships (Agnew et al., 1998; Lengieza et al., 2023). The component was informed by the meaning and emotion dimensions of the *Five Pathways* framework, while extending these through the introduction of biological interconnectedness as a reflective resource. This approach aligns with eco-existential perspectives suggesting that engagement with nature may support experiences of meaning, identity and belonging (Passmore & Howell, 2014).

Nature knowledge

Nature knowledge was included in response to research linking disconnection from nature to limited familiarity with natural environments and ecological illiteracy (Beery et al., 2023). The component provides a place-based narrative about local flora and fauna as well as the cultural and historical context of the environment, intended to support orientation to the local setting and foster place attachment as a resource for interpreting nature experiences.

In the guide, nature knowledge comprises two related elements. The first, familiarity with nature, refers to developing comfort and confidence in natural environments. To support this, the guide offers simple suggestions for where to direct attention and how to relate to the environment, enabling participation regardless of prior experience. The second element, place-specific knowledge, involves information about the particular setting in which the intervention takes place. It was included as a narrative anchor to encourage curiosity and reflection within the immediate setting, while situating the experience within a broader cultural, historical and social context.

Script development

After the three core components had been specified, they were translated into three audio tracks that together formed the audio guide. Each track operationalized the core components through a different thematic focus – air, trees or water – delivered as a narrative story about the immediate environment and guidance on how to engage with the surroundings. These themes were selected because they represent features commonly found across natural settings, thereby supporting transferability to other environments.

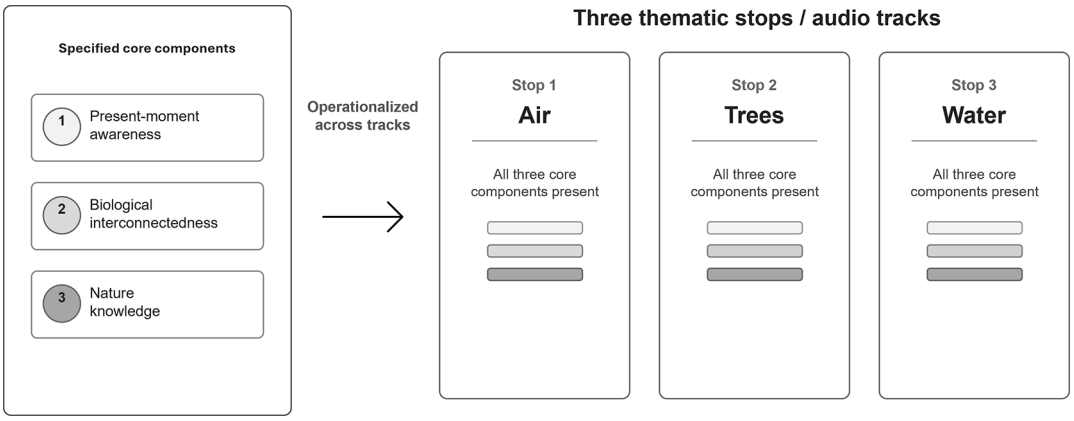


Figure 2. From core components to thematic stops. The three core components were first specified, then operationalized across three stop-specific audio tracks.

The script was drafted by the first author and reviewed by the second author with a focus on balancing theoretical integrity with practical usability and was subsequently revised based on feedback from the pilot study. The final guide consisted of an introduction and three audio tracks of approximately 7–9 minutes, delivered at three predefined stops along an accessible 1.2 km nature trail. The introduction and three tracks are described below.

Introduction

The introductory guidance welcomes participants to the nature stories and explains the structure of the walk, including the three stops and the chime. Its main function is to prepare participants for the guided format and support their transition into the guided nature experience.

> *Chime...*
> Welcome to the nature stories in the Arboretum. During the walk, you will stop three times to listen to a short nature story that awakens your senses and directs your attention to the nature around you. The beginning and end of each story are marked by this sound... *chime...*
> There will be pauses within the story, but it is not finished before you hear the tone and are guided to the next stop. You can now follow the path to the first stop.
> *Chime...*
> Now, let go of all instructions and simply float in a state of weightlessness, feeling connected to the air around you for a few minutes. When you hear the tone, it will be time to continue along the path.
> (Wait 3 minutes)

Figure 3. Manuscript for the introduction.

Air

The first nature guidance focuses on air. It encourages participants to engage with their surroundings through breathing and sensory noticing and to reflect on their experience of air. Participants are guided through a grounding exercise that incorporates bodily awareness, deep breathing, release of tension and attention to the environment, activating the Five Pathways contact pathway. Awareness of air is enhanced by directing participants' attention to the sensation of it flowing over their skin, the movement of the treetops and clouds, and the exchange of air between their body and the environment. Attention is directed to the sensory qualities of breathing, offering a frame for reflecting on the interdependence between body and natural environment.

Chime...
Welcome to the first stop, where you will listen to a nature story about the air around us. Find a comfortable place to sit and take a moment to calmly look around. (Pause)
When you are ready, take a few deep breaths, filling your lungs completely with air. Exhale slowly through your mouth. (Wait 25 seconds)
Now, turn your attention to your face. You may close your eyes if you like. Let the muscles in your face relax. Feel your forehead smooth out, release any tension around your eyes and relax your jaw. Allow a sense of calm and softness to spread. (Wait 10 seconds)
Now, shift your focus to your surroundings. Notice the sounds around you, both near and far. Try not to judge them as good or bad; they are all part of the experience in this moment. (Wait 1 minute)
If your eyes are closed, open them now. Let your gaze rest on a spot where you can see the sky. If it feels comfortable, you can lean back and look up. Observe the sky – perhaps it the colours are sifting, perhaps clouds are drifting by or perhaps the wind is moving through the treetops. (Wait 10 seconds)
You can also close your eyes and sense the air against your skin. Try moving your hands and arms slowly through the air and feel it around you. Let the experience of the sky above you and the air surrounding you gradually fill you with lightness. Just like our bodies, the air is made up of tiny atoms in constant motion. Through our breath, the air becomes part of us, absorbed into our cells. When you are ready, take a few deep breaths, filling your lungs completely. Exhale slowly through your mouth. Imagine yourself becoming just a little lighter with each breath. With every inhale, you are filled with lightness; with every exhale, you release tension. (Wait 40 seconds)
Now, let go of all instructions and simply float in a state of weightlessness, feeling connected to the air around you for a few minutes. When you hear the tone, it will be time to continue along the path. (Wait 3 minutes)

Figure 4. Manuscript for Air.

Trees

The second nature guidance is centred around trees. It begins with a short grounding instruction prompting bodily awareness and attention to a surrounding beech grove. A nature story highlights a specific variation of "twisted" beech trees that grow in the grove, emphasizing place-specific knowledge and diversity within species while subtly prompting opportunities for identification through metaphors. The story continues with a narrative about the sensory experiences associated with the budding leaves in spring, combining nature knowledge with the *Five Pathways.* Participants are guided through a visualization and breathing exercise that focuses on the process of photosynthesis and the exchange of nutrients in nature, drawing attention to interconnected and transformative processes in natural systems, which humans may consider themselves part of.

Chime...

Welcome to the second stop, where you will listen to a story about the trees around you. Find a comfortable place to sit and take a moment to calmly look around. (Wait 20 seconds)

In front of you, you see a gathering of beech trees. In dense forests, where sunlight barely reaches the ground, beech trees grow tall and straight, unfolding their leaves high up, close to the light. But when a beech tree has more space, its branches begin to spread just a few metres above the ground, forming a dome-like canopy around the trunk. Not all beech trees grow tall and straight. There can be great variation, even among trees from the same parent plant. The beech trees you see in front of you are part of the Arboretum's collection of twisted beeches. What makes these trees special is the way they twist into irregular, enchanting angles. Although they come from the same parent plant as the straight-growing beeches nearby, the genetic variation has only expressed itself in these twisted beeches, giving them their unique form. As spring approaches, something magical happens. The beech trees burst into leaves within just a few hours. Entire shoots, hidden within brown bud scales, unfold like small bouquets. On a calm, sunny day, you can hear the sound of the falling bud scales landing on the dry leaves beneath the trees, resembling the gentle sound of rain. The trees are part of a cycle connecting plants, animals and humans through photosynthesis, which transforms sunlight, water and carbon dioxide into sugar and oxygen. The oxygen produced by the trees is released into the air, and it is the same oxygen that humans and animals breathe to live.

Choose a tree and let your gaze rest on its crown. Try to imagine that you and the tree are breathing in unison. As the tree's leaves exhale fresh oxygen, you breathe it in. When you exhale carbon dioxide, the tree breathes it in. You are connected to one another. Now, let your gaze slowly follow the tree all the way down its trunk. (Pause) Imagine you can trace its roots deep into the earth, branching out far beyond the reach of its canopy. (Pause) The roots connect the tree to vital nutrients in the soil, absorbing them with the help of a vast network of fungi and bacteria living underground. We now know that trees can communicate with each other through the long threads of fungal networks. At this moment, you are sitting in the midst of nature's interconnected web, with branches above and roots below. Everything in nature – including ourselves – is connected and in constant transformation. Sit for a moment and let this sense of connection to the life around you take root within you. You don't need to do anything or think about anything in particular – just experience being part of this moment, right here and now. You

Figure 5. Manuscript for Trees.

Water

The third nature guidance focuses on water, and participants are seated in front of a lake. As in the other stories, they are first encouraged to heighten their bodily awareness and are then instructed to direct their attention toward sensory experiences related to the water and to reflect on similarities between the flow of water in the lake and within their body. The story contains more information about historical and biodiversity factors pertaining to the surrounding nature than the preceding stories, but this information is embedded in a reflective narrative to encourage participants' curiosity, emotional engagement and place attachment rather than convey scientific knowledge of the environment.

Chime...

Welcome to the third stop, where you will hear a story about the lake in front of you. Position yourself so you have a good view of the water and are seated comfortably. (Wait 5 seconds)

Take a moment to let your body settle. Feel the sensation of your body against your clothing and the support of the surface beneath you. Sit here for a while and calmly look around. (Wait 10 seconds) Let your gaze rest on the surface of the water and the sky above.

If there are small ripples on the water, allow your attention to drift with them. Perhaps the colour of the water changes with the sky and clouds, or a bird swims by. (Wait 1 minute)

Further to the right, the lake merges with Springdammen. In the old days, this was a meadow – there was no lake here, although it could get quite damp, especially in autumn. When Hirschholm Palace was built in the 18th century, a large amount of water was needed for its moats and the approximately 50 fountains in the palace gardens, located about a kilometre from here. To supply the water needed, a dam was constructed – the earthen embankment that can still be found at the Arboretum's main entrance. This created the lake that you now see before you. The palace and its fountains are long gone. The palace stood for less than 70 years, but in that time, it became the setting for the infamous affair between Queen Caroline Mathilde and Johann Friedrich Struensee, the royal physician to King Christian VII, an affair that ultimately led to the queen's exile and Struensee's execution. But the lake remains.

Ubberød Dam is not very deep, reaching only about 1.5 to 2 metres at its deepest point. However, the water is very clean, and the dense, healthy vegetation surrounding the lake provides excellent conditions for birdlife. As a result, the area is home to a wide variety of water birds, including the great crested grebe, coots, tufted ducks and swans. In early spring or late autumn, you might even be lucky enough to see an osprey passing through.

The nest box you see in the tree ahead is for a very special bird. If you are fortunate, you might spot common goldeneyes breeding here. While goldeneyes are common winter visitors, few actually breed in Denmark. Most of the breeding pairs are found here in North Zealand, near lakes surrounded by deciduous forest. Unlike most ducks, which nest on the ground, goldeneyes build their nests in tree cavities or use nesting boxes like this one. When the ducklings are ready to leave the nest, they are gently nudged out – so it is important that the box is positioned above the water.

The trees in front of you are bald cypresses. Like larch trees, they are among the few deciduous conifers. If you look closely at the base of the trees, you will notice small, knobby root structures emerging from the water's surface. These are called pneumatophores, or "breathing roots." Over time, they grow larger, almost resembling the tree's knees poking up above the water. They help

Figure 6. Manuscript for Water.

Audio production workflow

The decision to use an audio-based delivery format was based on evidence from smartphone-based "noticing nature" interventions, which have demonstrated that brief, structured prompts to actively notice nature can support improvements in mental health and affective outcomes, with some of these effects associated with increases in nature connection (McEwan et al., 2019). Key production decisions included the selection of the narrator's voice. A soft-spoken, clearly articulated female voice was selected based on its potential to promote listening comfort and speech intelligibility. Audio recordings were produced professionally in a recording studio, and a member of the research team oversaw formatting to ensure compatibility with smartphones. Aspects of the production were treated as part of the design. Specifically, voice, pacing, periods of silence and background audio were each intended to support engagement by promoting attentional orientation and facilitating emotional regulation. To increase usability, participants could access the guide in three ways:

single-ear headphone listening, loudspeaker mode at stops, and the option to pause or replay at any point. These options were included to address variation in participants' needs and preferences, in line with recommendations that complex interventions should allow flexibility and adaptation while maintaining core functions (Skivington et al., 2021).

Intervention site

The final intervention was designed for delivery in a forest-like arboretum covering approximately 25 hectares, located north of Copenhagen, Denmark. At the site, parking and accessible toilet facilities are available. The 1.2 km route consists mainly of firm, compacted soil and gravel, with a few sections of reduced accessibility due to woodchip surfacing and gentle inclines. The site had previously been shown to have high perceived restorativeness in relation to species richness, refuge and overall naturalness (Stigsdotter et al., 2017), and was considered well suited to nature-based experiences. Participants visited the route individually, but were permitted to be accompanied by a helper, family member or service dog. A research assistant was present throughout to provide practical support and monitor intervention fidelity.



Figure 7. The three stops, Photos by Ulrika K. Stigsdotter.

Stakeholder involvement and roles

Development involved an interdisciplinary research team covering nature-based health intervention research (psychologists and landscape architects), target-group contributors (adults with mobility impairments) in a pilot study, and project collaborators involved in site selection and route design (accessibility expert and park staff). The involvement of stakeholders had several functions consistent with complex intervention development: to assess the feasibility and acceptability of the guide, to identify aspects requiring refinement and to improve the intervention format in ways that supported its use in practice (Skivington et al., 2021).

Pilot study

To refine the engagement guide and route prior to full-scale testing, a pilot study at the intervention site was conducted in spring 2023. Participants were recruited using purposive and snowball sampling through a nature group within a disability organization. Inclusion criteria were that participants were adults with mobility impairments, were able to complete the route and were able to participate in an interview. The sample size was determined pragmatically because the pilot was intended to identify actionable issues of feasibility, acceptability, usability and fit, rather than to produce a standalone qualitative account. Five adults participated (aged 55–71 years; four women and one man), some using a wheelchair or walker. Written informed consent was obtained prior to participation.

Participants completed the route individually while accompanied by the first author, who provided assistance when needed in relation to accessibility, wayfinding or technical issues. Audio guidance was presented at three preselected stops. Informal observations made during route completion were used to inform follow-up questions in the interviews and to support practical refinement of the guide and route.

Immediately after completing the route, participants took part in an individual audio-recorded semi-structured interview conducted by the first author, either outdoors in a quiet area near the end of the route or indoors in an adjacent building. Interviews lasted 25–50 minutes and addressed participants' experiences of the stories, attention to the environment, usability of the audio format, route accessibility and suggestions for improvement. Interview recordings were transcribed clean verbatim and analysed using a thematic approach (Braun & Clarke, 2006) oriented toward intervention refinement. The translated interview guide is provided in the supplementary material.

The first author developed the guide, accompanied participants during the route, conducted the interviews and lead the analysis. This close involvement shaped contextual understanding of the pilot material and was addressed reflexively through discussions with the co-authors. The initial analysis was guided by two predefined areas, engagement with nature and interaction with technology, reflecting central design assumptions. Additional themes, nature literacy and moving between stops, were developed iteratively from the material. The resulting themes are presented below.

Engagement with nature

Participants described that their attention was often drawn to natural features during the walk. This spontaneous attentional orientation resembles processes described in theories of restorative environments, such as Attention Restoration Theory (Kaplan & Kaplan, 1989; Kaplan, 1995). Beyond this spontaneous attentional shift, the participants regarded the instructions as a novel mode of engaging with the environment, particularly through sensory grounding and present-moment orientation:

> *It is also a way to ground yourself. What you are hearing about is right in front of you. Thoughts may be fluttering all over, but you are able to ground yourself [through the story].* (Participant quote about present-moment instructions)

Nature literacy

Site-specific information about natural and historical elements of the environment played a noticeable role in how participants reflected on their experience. When talking about the walk and their relationship with nature, participants frequently referred to both existing knowledge and new information introduced in the stories. For several participants, concrete details and information appeared to provide an accessible entry point into reflection on the experience, particularly when more abstract aspects of nature connection were difficult to put into words:

> *I recognized it right away, that birdhouse … We would really like to have them [a certain species of birds] in our local area.* (Participant quote about nature knowledge and nature connection)

> *The stories about humans being connected with nature. I haven't thought much about that. I have mostly been interested in more technical details, I would call it, such as species identification, habitats and similar aspects … It was pleasant … Before I started the exercise, I was looking and listening for different species. I didn't think about the connection at all.* (Participant quote about nature knowledge and nature connection)

Interaction with technology

Participants accessed the stories by scanning a QR code that opened the audio files on their smartphones. Although introducing digital technology in nature may seem counterintuitive, given its association with reduced outdoor time and diminished attention to nature (Soga & Gaston, 2016; Webber et al., 2023), participants generally accepted the smartphone as a practical tool for delivering guidance. Their feedback aligned with recent work suggesting that digital media can support nature engagement by supporting discovery and familiarization with nature, increasing knowledge and experiencing environments from new perspectives (Webber et al., 2023). In this pilot, the audio format appeared to function in these ways by structuring attention and providing a scaffold for engaging with sensory and thematic elements of the environment. At the same time, participants noted disruptions linked to the audio format. Some experienced uncertainty during longer pauses in the stories, prompting them to check their phones and momentarily break immersion. Others found that wearing earphones interfered with listening to ambient sounds, especially when the task involved listening to birds:

> *They [the stories] were actually nice, except for the one where you are encouraged to listen to the birds. It's challenging to do so with the headphones on. Which type of bird am I supposed to listen to? I had to remove the headphones, and as a result, I missed parts of the story. There should be an option to pause, remove the headphones, and listen.* (Participant quote about interaction with technology)

Moving between stops

Participants differed in how they navigated the route between the predefined stops. Several moved quickly through the sections in between, treating the stops as the primary points of engagement. For some, the fixed stops unintentionally highlighted only selected features of the environment, which led to a sense that the spaces between them were less meaningful or overlooked. One participant explicitly noted a desire for more guidance or information during these transitions.

> *I think you should include more descriptions of nature, especially the smaller details. We passed by a place with a lot of wild garlic… That's what you can smell here.* (Participant quote about moving between stops)

This theme illustrates how the spatial structure of the route shaped participants' attention. While the predefined stops concentrated engagement at specific points, they also appeared, for some participants, to restrict attention to those locations at the expense of the wider environment between stops.

Taken together, the themes suggested that the guide was acceptable and potentially supportive of engagement, but also that its effects depended on the interaction between content, delivery format and route structure.

Refinement of the guide and route design

These findings informed iterative refinement of the guide and route design. They supported retaining site-specific nature knowledge as a core component, while pointing to a need for adjusting the audio format, making decisions regarding the handling of transitions between stops and changing part of the route.

The pilot indicated that even though participants generally accepted the smartphone as a delivery tool, two key design tensions were identified:

1. *Pause uncertainty and device checking*

The pilot revealed moments of uncertainty during longer pauses in the audio guide, which led some participants to check their phones and temporarily break immersion. To address this, a low-level background birdsong was added to signal intentional pauses and act as implementation support. Birdsong was chosen as it has been associated with pleasurable experiences in everyday nature experiences (Ratcliffe et al., 2013; McEwan et al., 2020).

2. *Sensory interference via headphones*

Participants noted that the headphones at times interfered with their ability to hear ambient sounds such as birdsong when prompts invited attention to nature sounds, revealing a structural conflict between guided audio delivery and the ambient sound environment. This led to adaptations including single-ear listening, pausing the guide and loudspeaker use.

Beyond these audio-related tensions, two additional design considerations were identified:

1. *Route structure and pacing*

Pilot participants varied in their pace and their engagement with the surroundings between predefined stops. Although some participants expressed a desire for additional guidance during these transitions, no guidance was incorporated between stops. This decision reflected practical and conceptual considerations: it would require a high degree of control over environmental features subject to seasonal change and ongoing site maintenance, and it would limit participants' freedom to remove their earphones and engage spontaneously with the environment between guided stops.

2. *Environmental setting and contextual fit*

Feedback from the pilot suggested that the setting was broadly suitable, as participants were able to engage with both sensory and thematic elements of the stories. However, the route was adjusted slightly to optimize physical accessibility, which also required one story to be rewritten to fit the revised route.

Programme theory

As part of the complex intervention development process, a programme theory and logic model was developed to specify how the engagement guide is expected to support experiences during accessible nature visits. The theory proposes that, when participants can access a setting that affords meaningful participation, and when there is a good fit between the user, the guide and the environment, the guide may support processes such as present-moment awareness, reflective engagement, emotion regulation and different ways of relating to nature. Although nature connection was an important starting point in the guide' development, this framing proved too narrow as the intervention developed beyond a sole focus on nature connection. In the final programme theory, nature connection is therefore retained as an important conceptual influence and possible relational pathway, but not as the guide's primary or necessary mechanism of action. These engagement-related processes may contribute to short-term restoration and wellbeing-related experiences and, for some participants, to experiences of connection or resonance**.** When fit is poor, however, the guidance may introduce burden, irritation, or disengagement, thereby reducing the quality and supportive potential of the experience.

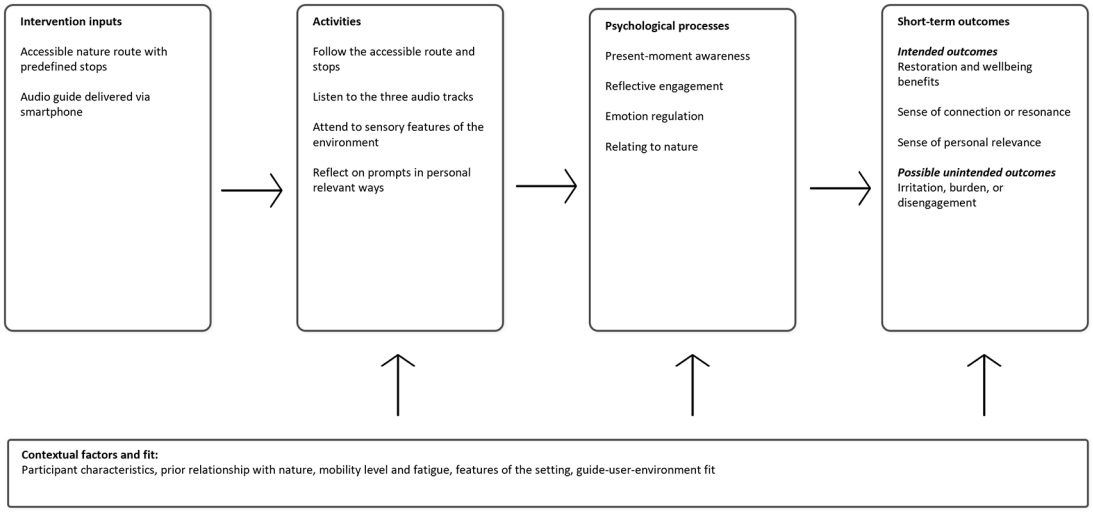


Figure 8. Logic model of the engagement guide. The model presents hypothesized, context-dependent relations. Arrows indicate plausible links and do not imply fixed or linear causal pathways.

Discussion

This article contributes to nature-based health intervention research by providing a theory- and data-informed account of how a place-based audio guide was developed to support structured engagement with nature among people with mobility impairments. This is relevant in a field where key intervention mechanisms and active components are increasingly recognized as important, yet often remain insufficiently specified (Macaulay et al., 2025; Madsen et al., 2026). The development and refinement process showed that both the content and the delivery format of the guide shaped how participants engaged with the environment. Pilot feedback suggested that the audio format could both support and constrain attention, at times competing with direct sensory contact with the surroundings. This indicates that features such as pauses, background sound and listening modes should be treated as intervention elements rather than as minor delivery details.

Site-specific nature knowledge and prompts for present-moment attention appeared to offer concrete and accessible ways of engaging with the environment, underscoring that guidance can shape how participants attend to, interpret and interact with natural settings. Building on previous research showing that the potential benefits of nature contact depend not only on contact itself but also on the quality of that contact (McEwan et al., 2020; Richardson et al., 2022), the present findings suggest that the guide may help structure the quality of such engagement. However, engagement was shaped not only by the audio guide, but also by participants' prior experiences and preferences and by the specific environment in which it was used. This is particularly relevant in the present target group, as physical access to a natural setting does not in itself ensure equal access to meaningful engagement with that environment (Bekke-Hansen et al., 2023; Corazon et al., 2019). The study therefore supports an understanding of accessibility that extends beyond physical entry into a setting and includes the conditions that make meaningful engagement possible in practice. In this sense, the audio guide may be understood as an attempt to support experiential access as well as physical access. These findings also have implications for how the intervention is conceptualized and adapted across settings. Rather than being defined primarily by its fixed

place-specific content, the audio guide may be better understood in terms of its core functions: orienting attention, supporting calm engagement and offering interpretive prompts that may strengthen meaningful contact with the surroundings. This distinction between core functions and adaptable forms is consistent with current guidance on complex interventions and may be particularly important in nature-based intervention research, where context is part of the intervention rather than a neutral backdrop. This should also be taken into account when considering transferability to other settings.

A strength of the development process is the theory-informed and iterative design, active stakeholder involvement, explicit attention to accessibility and incorporation of real-world implementation feedback. Together, these features strengthen the transparency of the intervention development process and help make the intervention's assumptions, intended functions and design decisions more analytically visible. However, several limitations should also be acknowledged. Development work can identify issues of fit, feasibility, acceptability and conceptual coherence, but it cannot establish whether the intervention is effective, which components are necessary or sufficient for outcomes, or which mechanisms are most important for whom. In addition, transferability to other settings and populations remains to be examined.

The audio guide developed in this study now provides a basis for subsequent evaluation using both quantitative and qualitative approaches. In the next phase, this will take place through an ongoing randomized controlled trial, in which the audio guide serves as the guided intervention condition, allowing examination of effects on nature connection, health and wellbeing (Bekke-Hansen et al., in review). Alongside the trial, the programme theory will guide an embedded qualitative investigation of how the intervention is implemented and experienced in practice. This work may help clarify how the audio guide functions, which elements should be treated as core to fidelity, and which may be adapted across settings without undermining the guide's intended purpose. More broadly, the study highlights the value of considering engagement itself as a focus of intervention development in nature-based interventions.

Conclusion

This development study shows how a place-based audio guide can be specified and refined to support structured engagement with nature among people with mobility impairments. It highlights the importance of designing not only for access to nature, but also for the quality of engagement once access is achieved. More broadly, it illustrates how engagement can be treated as a developmental target in nature-based interventions. The audio guide is currently being evaluated in an ongoing larger-scale study (Bekke-Hansen et al., in review).

AI declaration

During the preparation of this article, the authors used Microsoft Copilot Enterprise to search for and structure information, and to translate, proofread and rephrase wording for clarity and readability. The tool was not used to generate scientific insights. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the article content.

Ethics approval and consent to participate

The pilot study was not submitted to a Danish research ethics committee because it was conducted as intervention-development and feasibility work and did not aim to generate new knowledge about disease, treatment, prevention, rehabilitation, or biological, physiological, or psychological processes. The pilot examined acceptability, usability, route accessibility, and contextual fit of the audio guide before later health-related evaluation. The processing of personal data for the pilot was registered at the University of Copenhagen's registry of research projects involving personal data (reference no. 3163798/4242). The subsequent main project was approved by the Danish Regional Committee on Health Research Ethics (record no. H-23077317), and the processing of personal data was approved by the Faculty Secretariat, Faculty of SCIENCE, University of Copenhagen, which confirmed compliance with the General Data Protection Regulation (GDPR; Regulation (EU) 2016/679) (record no. 514–0487/24–5000).

All participants received written and oral information and provided written informed consent before participation. Participants were informed of their right to withdraw from the study at any time without penalty. No financial compensation was provided. Personal data were handled confidentially and stored securely.

Data availability statement

The data underlying this paper are not publicly available. The pilot study data consist of qualitative interview and feasibility material from human participants and include detailed contextual accounts that may be indirectly identifiable even after de-identification. Public sharing of the raw dataset would therefore be inconsistent with the consent conditions and with the confidential handling of personal data described to participants.

As described in the Ethics approval and consent to participate statement, the pilot study was registered at the University of Copenhagen's registry of research projects involving personal data (reference no. 3163798/4242), and all participants provided written informed consent before participation.

Access to the raw pilot study data is restricted for ethical and data protection reasons. Requests for access to de-identified excerpts or limited data materials for purposes of verification may be directed to Stine Bekke-Hansen, University of Copenhagen, sbh@ign.ku.dk. Requests will be considered on a case-by-case basis and access will only be granted where compatible with the original participant consent, confidentiality obligations, University of Copenhagen data protection requirements, and applicable GDPR rules. Access may require a justified research purpose, appropriate institutional approval where relevant, and a data sharing or data processing agreement.

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The authors would like to express their gratitude to colleague Patrik Karlsson Nyed, Academic Research Staff, whose critical engagement supported the development of the intervention.

The authors further thank Ole Rasmus Byrgesen, Arboretum supervisor, for sharing his expertise on the Arboretum's unique collection of trees and shrubs, and Rikke Stener Nielsen, who ensured that the intervention trail was clearly marked and accessible.

We acknowledge the 15. Juni Fonden for facilitating access to the Arboretum, which enabled the intervention.

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?

Christopher Barnes
University of Derby, Derby, UK

I found the manuscript to present an interesting and important piece of work that aimed to develop and describe a place-based audio guide for accessible nature visits and to refine the intervention prior to further evaluation. Most sections were well written and clearly communicated, although some of the introductory and analytical sections were relatively brief. I particularly valued the strength-based approach adopted by the authors, as described in the Reporting and Methodological Standards subsection, as well as the broad theoretical framework used to inform the development of the audio guides.

However, while I found much of the methodological underpinning to be sound and sufficiently comprehensive, I do have some reservations regarding the pilot study and subsequent qualitative analysis.

First, while participant numbers do not need to be substantial in qualitative research, nor equally balanced by gender, recruiting only five participants is still somewhat limited. Braun and Clarke typically suggest broad estimates of around 15 participants when conducting reflexive thematic analysis. However, it is acknowledged that smaller samples may be appropriate when the research team considers that sufficient information power has been achieved. The authors do not appear to indicate that this was the case, although doing so could help justify the small sample size.

Second, the authors do not provide sufficient information regarding their methodological and analytical positioning, including their epistemological and ontological assumptions. I would have welcomed more detail on whether the analysis adopted an inductive or deductive approach and whether it was informed by a realist, constructionist, or critical realist position.

Third, the qualitative analysis appears somewhat diluted, perhaps because a substantial proportion of the findings section is devoted to describing the development of the guides rather than analysing participants' experiences and exploring their meaning. As a result, the analysis feels relatively light, and I believe the section would benefit from rebalancing to provide a more comprehensive representation of participant perspectives. While I recognise the value of detailing

the development process and the steps taken to create the guides, I felt that this aspect was given disproportionate attention relative to the analysis.

Fourth, related to the previous point, the presentation and evaluation of participant voices was also limited and, in some instances, did not fully align with qualitative reporting conventions. Participant extracts were relatively sparse, were not linked to identifiable participants, and were not explored in sufficient depth to illuminate the experiential aspects of engaging with the intervention and audio guides. Although I appreciate that the study has a strong feasibility focus, more could have been done to examine how participants' experiences related to relevant theory and existing research.

Finally, participant characteristics and a summary themes table were either absent or not presented in sufficient detail. While I recognise that the paper is not solely focused on the qualitative component of the pilot study, the inclusion of this information would strengthen the manuscript. This would be particularly beneficial if accompanied by a more detailed qualitative analysis.

Is the rationale for developing the new method (or application) clearly explained?
Yes

Is the description of the method technically sound?
Yes

Are sufficient details provided to allow replication of the method development and its use by others?
Partly

If any results are presented, are all the source data underlying the results available to ensure full reproducibility?
Partly

Are the conclusions about the method and its performance adequately supported by the findings presented in the article?
Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Nature Connectedness, Nature-Based intervention, Qualitative and quantitative analytical expertise, Feasibility trials.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Reviewer Report 04 September 2026

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Donald Rakow
Cornell University, Ithaca, USA

I admire the authors' goal of increasing access to nature for individuals with mobility limitations. I also greatly admire previous research conducted by Dr. Stigsdotter. So, while I read this piece with great admiration, I cannot recommend it for indexing for a number of reasons, which I enumerate below.

First, the article is based on a pilot study of only five participants using wheelchairs or walkers. Although the feedback from these individuals helped to refine the approach and script for a full-scale study, the small number of participants, along with the absence of a control group, result in this pilot lacking meaningful results.

Second, while the intended audience for these audio guides is physically disabled persons, there is nothing obvious about the audio guides that would make them particularly appealing to such individuals. Why would audio guides not be of equal appeal to fully abled folks?

Third, I question the hypothesis that audio guides will enhance individuals' attention, connection to nature, sense of place, or knowledge of nature. As is stated in the article, "participants noted disruptions linked to the audio format." Some participants also stated that "the fixed stops unintentionally highlighted only selected features of the environment." I would suggest that a more immersive approach would be to engage participants in forest bathing or forest therapy, where a human guide provides invitations that individuals can then follow along the entirety of a nature-filled path.

Fourth, for the audio guide format to demonstrate that it provides benefits for the intended audience, the authors would need to lead a study with a much larger number of subjects, an equally sized control group, and a clearer set of objectives. These are apparently all part of a larger study by Bekke-Hansen et al. currently in review.

Finally, given the polyglot of languages spoken in European countries, I feel that recording these audios guides in a single language would, by necessity, limit their utility. I am thinking particularly of immigrants who don't necessarily speak the native language.

Is the rationale for developing the new method (or application) clearly explained?
Partly

Is the description of the method technically sound?
Yes

Are sufficient details provided to allow replication of the method development and its use by others?

Yes

If any results are presented, are all the source data underlying the results available to ensure full reproducibility?
Partly

Are the conclusions about the method and its performance adequately supported by the findings presented in the article?
No

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Human benefits of time in nature, particularly for college-aged youth, Ways in which gardening can impact human well-being, The history and community impact of botanic gardens.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

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?

Matthew Owens
University of Exeter, Exeter, UK

Developing a place-based audio guide to support nature engagement: Intervention content and programme theory

This is a valuable piece of work detailing how nature engagement can be supported for those with mobility impairments. The latter point should be included in the title, so it is clearer to the readers that the pilot recruited individuals via a nature group in a disability organization.

While the introduction sufficiently laid out the background to the research, there is existing supporting research that could be referenced on the effect of nature-based meditation and guided imagery (e.g Ref 1)
This is slightly different work of course but is related.
It wasn't clear to me why the audio (with stops) was particularly important for those with mobility issues and not useful more generally. In other words, why was this approach thought to be particularly important for mobility issues and less so for those without such problems? Why could standard approaches not be used for those with mobility issues? This rationale wasn't made clear

enough and should be discussed.

A number of limitations should be raised or expanded upon. These are detailed below.
The sample size for this study was very small, even for a pilot study. Furthermore, gender was not balanced (4 females, 1 male). While this is ok for an initial exploratory piece of work, the limitation should be acknowledged.
Relatedly, it is hard to tell whether the themes that have been derived through interviews are exhaustive. Saturation may or may not be achieved with more or longer interviews. Future work could assess. The wide range of interview length should also be discussed (25-50 minutes). Standardising would have been better to avoid undue weighting from individuals.
It was surprising that no quantitative data at all was collected, especially given that one of the aims of the study was to assess acceptability. This data should be used to support the full trial before it started ideally. I couldn't see the supplementary materials from my copy. These should be clearly linked.

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Is the rationale for developing the new method (or application) clearly explained?

Yes

Is the description of the method technically sound?

Yes

Are sufficient details provided to allow replication of the method development and its use by others?

Partly

If any results are presented, are all the source data underlying the results available to ensure full reproducibility?

No

Are the conclusions about the method and its performance adequately supported by the findings presented in the article?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Nature-based interventions.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.